Manual

Machining Center / Pool of Orders

BDE-BEA 8.2

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Contents

1 Overview: Machining Center/Pool of Orders

Purpose

Using this component, you can illustrate issues of a "machining center/pool of orders".

Implementation notes

This function package is used if you have machines or machining centers where different items/orders are processed simultaneously.

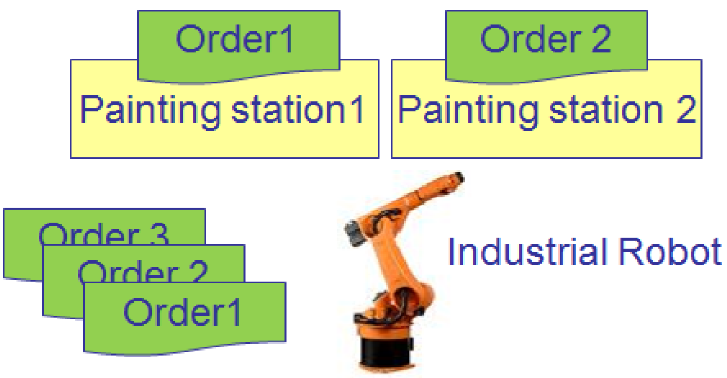
In a machining center several operations are logged on to a machine (= machining center) at the same time. As several operations are processed sequentially at a machine, they need to be differentiated to post the duration and labor utilization only to the active operation. In HYDRA MDE (Machine Data Collection) produced quantities are only posted onto active operations.

In the classical sense of a machining center, e.g. a robot works on several tables with different orders at the same time. In a machining center several orders can be processed alternately at a machine or workstation.

This example is to explain how machining centers can be used. A machining center has two painting stations that are fed by a robot. Painting station 1 uses the color blue and painting station 2 uses the color green. Different articles are produced (item 1 and item 2). One operation or one production order is required for each article/item to be produced. This example has two production orders:

	Blue	Green
Article	Item 1	Item 2

These two articles can be produced and painted at the same time. The items can be processed in different order or with different quantities at the robot.



Integration

Two different connection or collection options are possible:

1. Interactive activation of an operation (from the list of running operations) at the terminal. Downtime monitoring is realized by machine interface/UMPS or manual notification.
2. Recording of the events "order start" and "order end" using an interface. Provided that machines are connected correspondingly, information can be received from the machine or machine control to automate the change of operations.

Functions

- Functions to map specific requirements when working with a pool of orders at machining centers
- Configuration of machines and workplaces as machining center including pool of orders
- Login of several operations as pool of orders and login of people to the Windows terminal
- Recording of the events order start and order end, i.e. posting of the actual time when processing of an operation has started or ended
- Automatic registration of events by data taken over from machine controls (PLC) using an interface or manual registration by inputting events at the shop floor terminal
- Downtime monitoring at the Windows terminal including alternative downtime monitoring using machine interfaces or manual input
- Direct posting of recorded quantities and production times onto the corresponding operation
- Proportionate posting of downtimes and malfunction periods onto the registered operations

2 Machining Center / Order Pool

Purpose

You can use the functionality "Machining center/Order pool" on the AIP to process the operations, which are logged on in parallel, in a specified sequence.

Requirements

If you use AIP 8.2, you require at least version AIP 8.2.1.26 or service pack 13 (2018).

If the machine data collection is run via a stand-alone PCC ("central MDE"), you require at least service pack 13 (2018) and AIP 8.2.

Check in the Dynamic dialog configuration, that the dialogs NC_AN and NC_AB are existing and activated. If the dynamic dialog NC_AN and NC_AB are not available, proceed as described in chapter *Load dynamic dialogs*.

If required, execute the configurations described in chapter *AIP configuration* .

Activation

You enable the functionality "Machining center" on the MOC in the machine/workplace configuration:

Go to: *Master data* → *Workplace/resource configuration* → Tab *Workplace configuration* → *Workplace category=J <Machining center>*

The functionality "Machining center" includes the specific posting of durations. For this reason, you must set the option *Posting of machine time with operations logged on simultaneously* to "N" (also included in the tab workplace configuration).

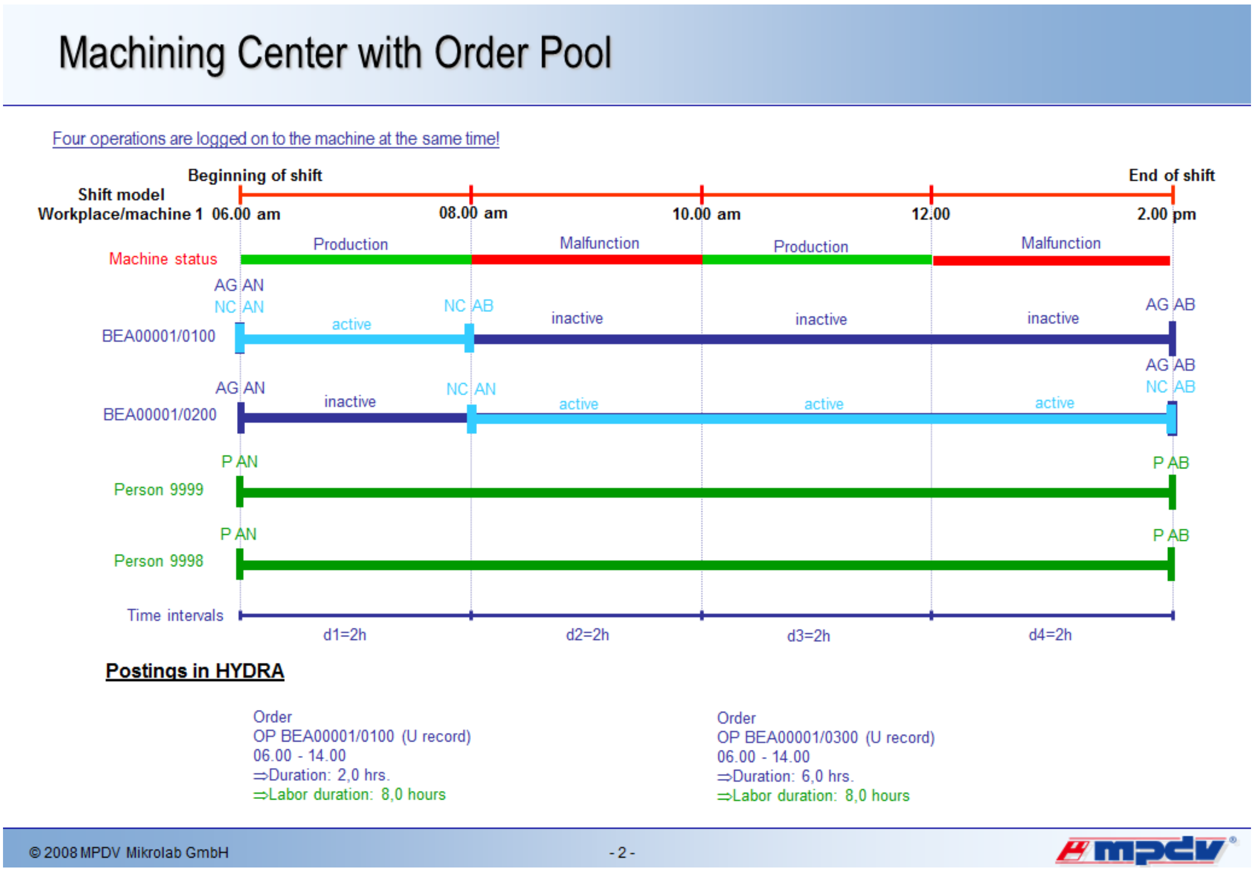
This functionality is also available on the AIP 8.2, if the PCC is run stand-alone ("central MDE").

Posting procedure and posting

The operator logs on the different operations to the respective machine. Further persons can optionally log on.

One operation is processed at a point in time X. If the machine is in status "Production", the duration and the complete labor utilization is posted to the active OP. If no operation is active, the system does not record any duration for an order. In case of a machine failure, the duration and the labor utilization are proportionally posted to all operations.

If the system records automatic quantities for this machine, the quantities and the production times are posted the same way: the automatic quantities are only posted to the active operation and for the persons logged on to this operation. Manual quantities are assigned to the operation to which the quantities are posted.

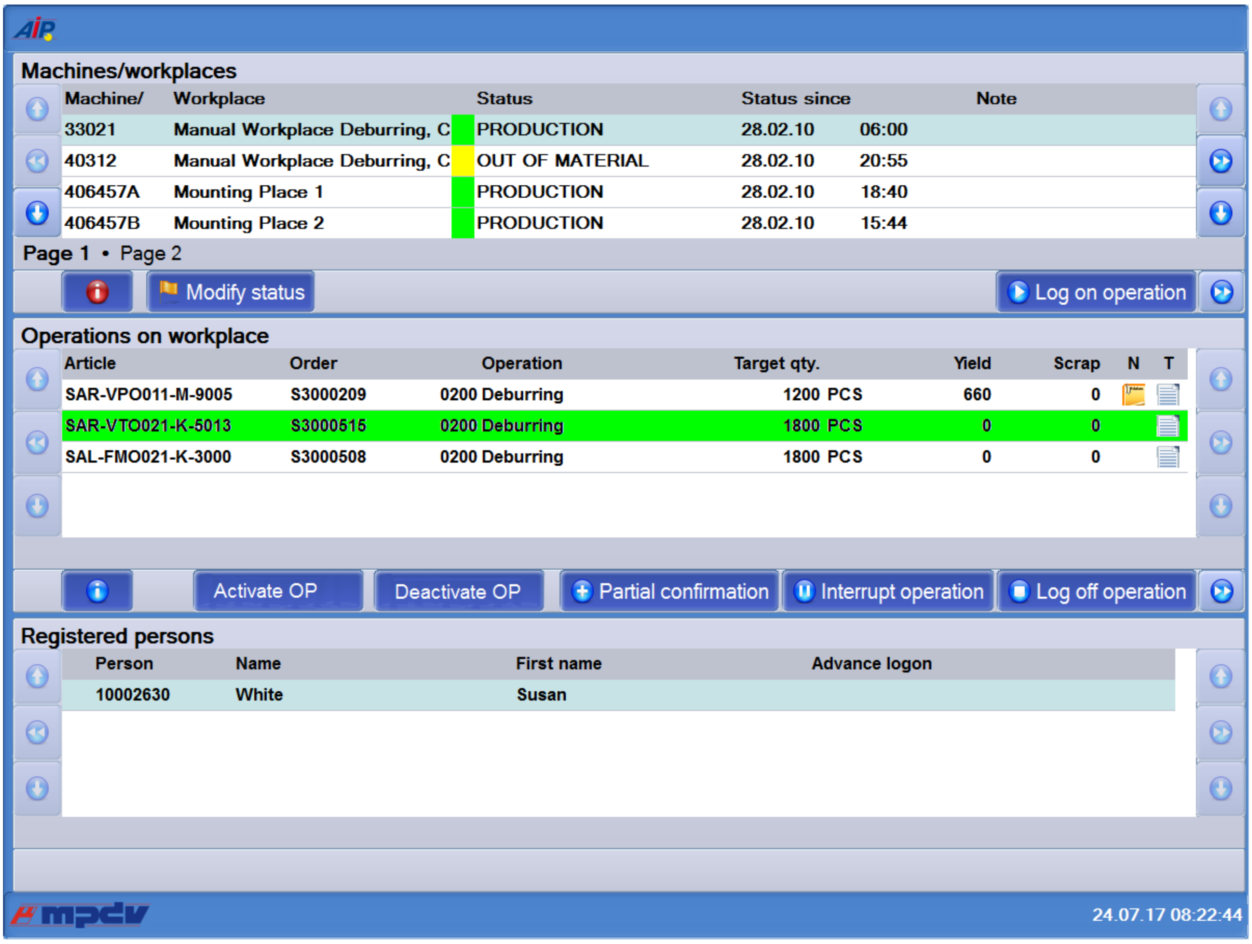


Also a HYDRA group workplace can be a machining center.

You cannot create merged operations for a machining center, because the production times and downtimes are always posted proportionally. This processing of postings contradicts the processing of postings for a machining center.

The posting events "Activate operation" and "Deactivate operation" are displayed in the event maintenance (events NC_AN, NC_AB).

Functions on the terminal AIP 8.1



The list "Operations on workplace" of the main view shows all operations logged on. If the order list is configured accordingly, the operation currently active is colored (green).

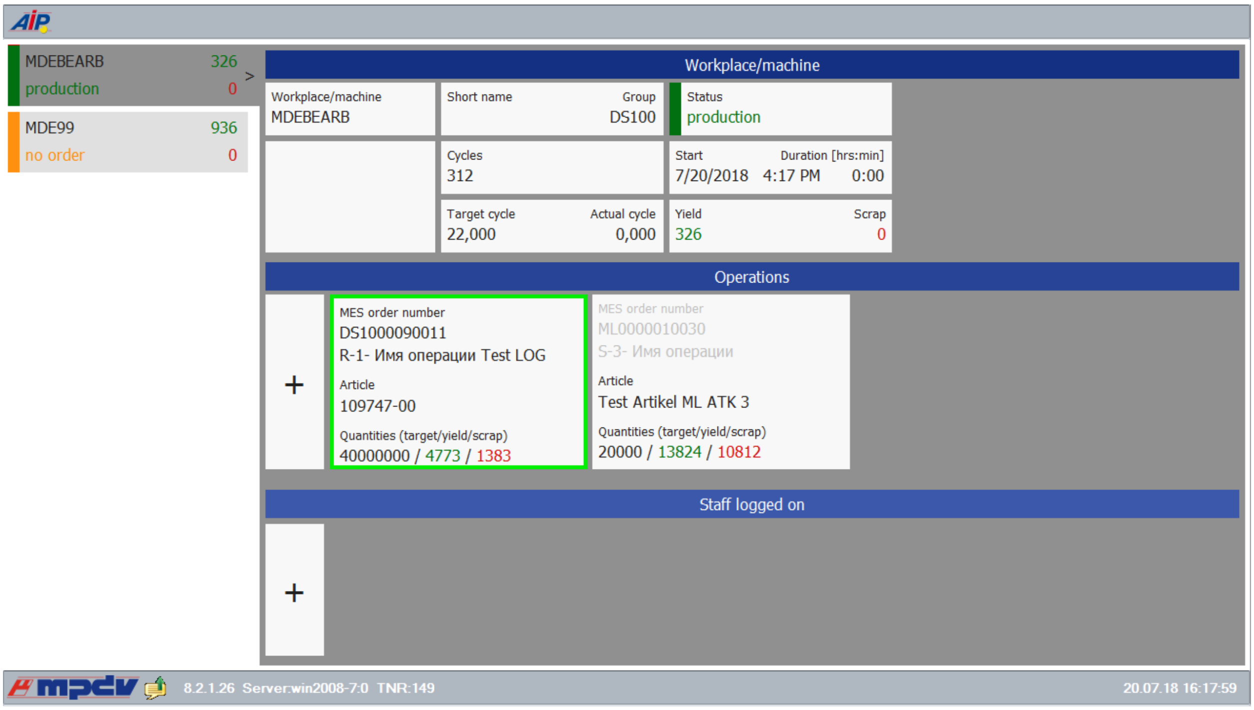
In the dialog "Activate OP", the operation selected in the main view is used and preassigned.

In the dialog "Deactivate OP", the active operation is automatically set.

Functions on the terminal AIP 8.2

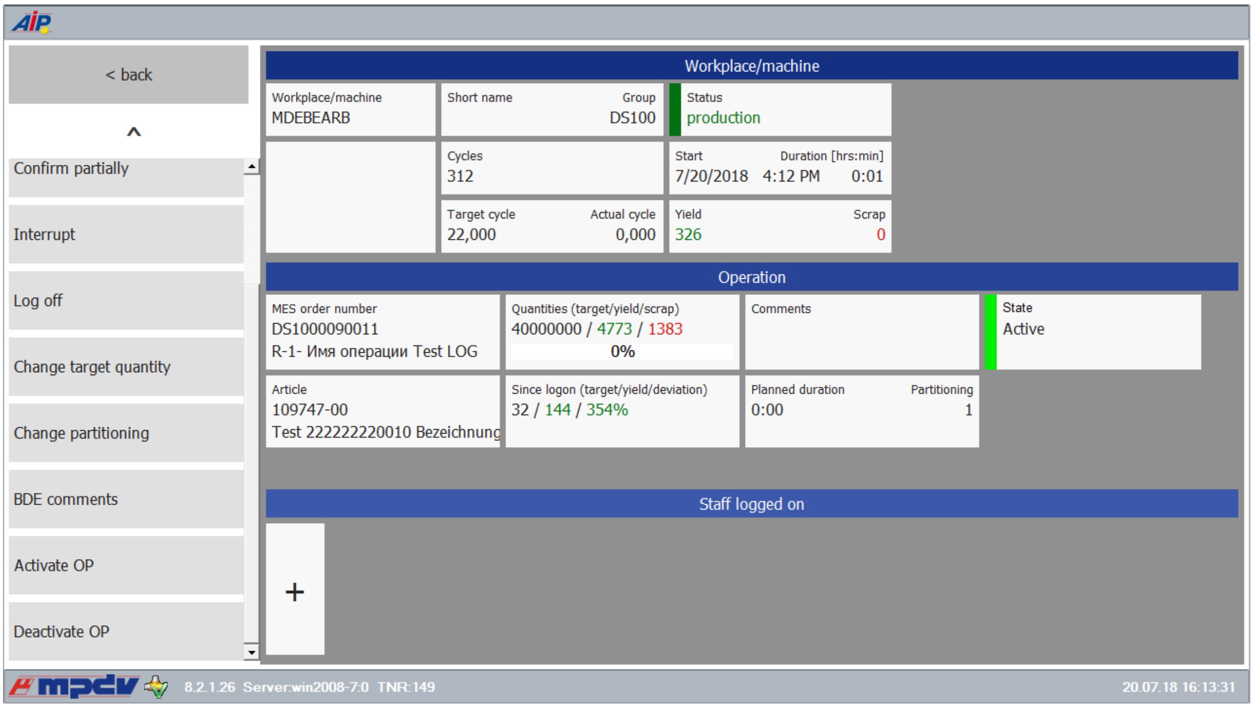
Main view

The list "Operations on workplace" of the main view shows all operations logged on. A green frame highlights the active operation. The MES order number of the inactive operations is displayed in gray.



Detail view of the operation

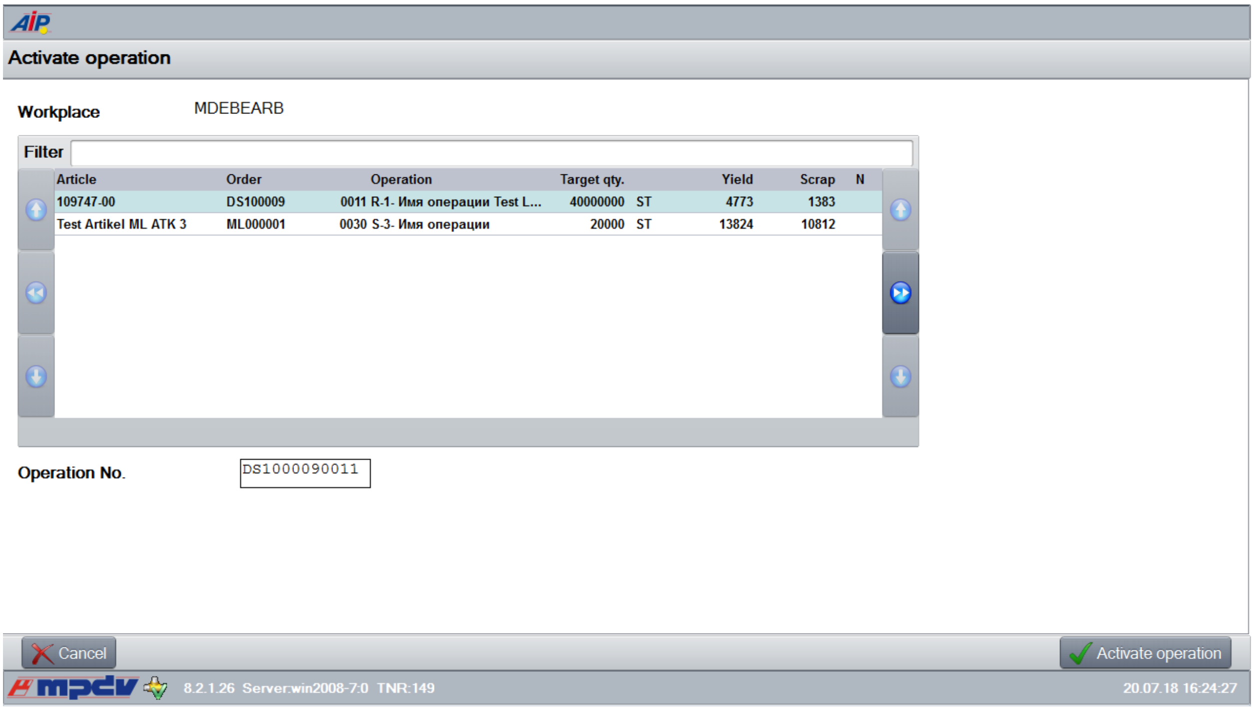
To activate or deactivate an operation, just click on the operation. In the detail view of the operation, the section "Operation" shows if the operation is active (green) or inactive (gray).



You can use the buttons on the left hand side to "Activate operation" or "Deactivate operation".

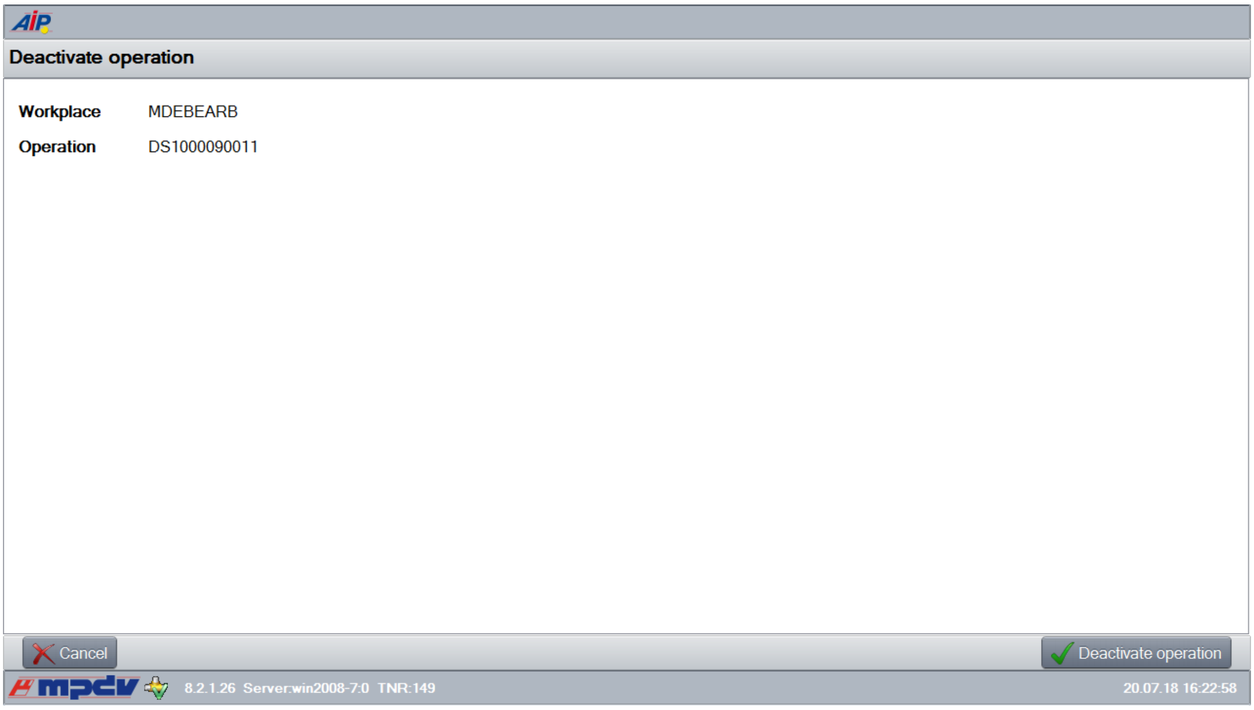
Dialog "Activate operation"

In the dialog "Activate operation", the selected workplace is pre-assigned. You can select the operation in the list that you want to activate. If an operation is currently active, this operation is shown in green until you select the row of this operation.



Dialog "Deactivate operation"

In the dialog "Deactivate operation", the selected workplace and the selected operation are pre-assigned.



Machine connection

To connect a machine, you must coordinate all details with the customer or the machine manufacturer. A subsequent customization is required. The following details must be discussed:

Details of technical communication.

- Examples: OPC-UA, File-Interface, UMPS, etc.
- Is the PCC run in stand-alone operation? (Note: only possible if you use AIP 8.2)

Details of logical communication.

- Can the machine control provide the data listed in the table below? If no:
  - How can the system identify this information?
  - Do you have to make extensions in the data collection dialogs?
- Which format does the data provided have?
- Does the machine control send a command to activate an operation **and** then to deactivate an operation or does the machine control only send a command to activate an operation?

Commands to activate or deactivate an operation

Field ID		Description
DLG		Dialog/record ID: always: „NC_AN“ – activate OP. „NC_AB“ – deactivate OP.
MNR		Workplace/machine number according to configuration
ANR		MES order number (order/OP)

Beforehand, you must manually log on the operations on the terminal.

Identification of the MES order number

If the machine control cannot pass the MES order number, a specific customization is required to add the MES order number (ANR) to the command to activate or deactivate an operation.

Load dynamic dialogs

Execute the following activities on the HYDRA server in the system directory.

1. Save the existing dialog configuration:

   UNIX systems (run at the server prompt in the HYDRA directory):
   `hydlgcfg.out DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg`

   Windows systems (Execute in a DOS window in the HYDRA directory):
   `hydlgcfg.exe DLGCFG.TYP=%% DLGCFG.DLGUSR=%% DLGCFG.DLG=%% DATEI=dialog.dlg`
2. Now load the new dialog configurations using the following command:

UNIX systems:
hymw.out -u9999 -b db_sql/aip_bde_bea.dlg

Windows systems:
hymw.exe -u9999 -b db_sql\aip_bde_bea.dlg

Please note:
If UNIX systems are used, you must enter the file name in lower case letters, as HYDRA Install converts all file names into lower case letters during installation.

3. Activate the new dynamic dialogs:

   UNIX systems:
   `hydialog.scr AIPTNR 0`

   Windows systems:
   `sh.exe hydialog.scr AIPTNR 0`

   Note:
   This command activates the default dialogs. If terminal or terminal group-specific dialogs are used on the system, they must be customized by an MPDV consultant.

4. Execute configurations described in chapter "Activation" or the "Configuration AIP". If terminal or terminal group-specific dialogs are used on the system, the settings should ideally be made with the support of an MPDV consultant.

5. Start the terminal and test the function.

AIP configuration

Display of the active operation on the terminal AIP 8.1 (if required)

Older installations of the AIP 8.1 can still include the following configuration. Make the following settings in the ctaiplay.ini to highlight the currently active operation on the terminal:

```
[order list]
…
GRID_COLOR=clSilver
GRID_BACKGROUND=clWhite
EXAMINE_SCANEXPR1=AKTIV=J
EXAMINE_SCANEXPR2=AKTIV=N
EXAMINE_SCANCOLOR1=clBlack
EXAMINE_SCANBKEXPR1=AKTIV=J
EXAMINE_SCANBKEXPR2=AKTIV=N
EXAMINE_SCANBKCOLOR1=clLime
EXAMINE_SCANBKCOLOR2=clWhite
…
```

Configuration of the buttons on the terminal AIP 8.1 (if required)

Older installations of the AIP 8.1 can still include the following configuration. Make the following settings in the ctaipbut.ini to show the buttons Activate and Deactivate on the terminal. You can position the buttons anywhere as long as the existing buttons are respected. You must ensure that the buttons are entered in the required section of the file ctaipbut.ini and that numbers are consecutive.

Example:

```
[ANR-ALL-Page4]
1=NC_AN,L,Activate OP
2=NC_AB,L,Deactivate OP
```

Automatic deactivation of active operation
Parameter in ctaip.ini:

```
[System]
NC_AUTO_DEACTIVATE=Y
```

Use this parameter to disable the plausibility function, which ensures that only one operation can be active. At the same time, the following automatic is active when you activate an operation:

The currently (previously) active operation is first deactivated. Then the new operation is activated. Note: The system does not perform these processing steps in one transaction. The system also deactivates the previously active operation if the new operation cannot be activated because of a plausibility error.